

Scientific Methods

AirOne V7.1 ships **37 deterministic scientific analyses** across **9 categories**, implemented in `src/scientific/` and registered in `src/scientific/registry.py`. Every analysis returns a `ScientificResult` with a value, unit, explicit **uncertainty**, and the assumptions it relied on.

Scientific-honesty rules

These rules are enforced in the analysis code and are non-negotiable:

1. **Correlation is never reported as causation.** Correlation analyses (e.g. `radiation_altitude_correlation`, `correlation_matrix`) report the coefficient and sample size and explicitly label the relationship as correlation only.
2. **Uncertainty is always propagated.** No analysis returns a point value without an uncertainty (or an explicit statement that uncertainty is not yet quantifiable, in which case the result is marked accordingly).
3. **Insufficient data yields an explicit “cannot compute”, never a guess.** An analysis with too few valid samples returns an unavailable/insufficient result rather than a fabricated number.
4. **Invalid inputs are excluded, not imputed.** Analyses consume only valid measurements; missing values are never zero-filled.
5. **Simulated inputs stay labelled.** Results computed from `SIMULATED` data inherit that provenance.

Categories and analyses

Atmospheric (10)

`atmospheric_density`, `atmospheric_stability`, `boundary_layer_height`, `dew_point`, `hypsometric_altitude`, `lapse_rate`, `pressure_gradient`, `temperature_gradient`, `temperature_inversion`, `vertical_profile`.

Core relations used include the barometric/hypsometric equation for pressure-derived altitude, the environmental lapse rate (dT/dz), the Magnus/Arden-Buck formula for dew point, and the ideal-gas relation for air density. Stability and inversion detection compare the measured lapse rate to the dry adiabatic lapse rate.

Descent (4)

`descent_rate`, `drag_coefficient`, `landing_detection`, `terminal_velocity`.

Descent rate is the time derivative of altitude; terminal velocity and the effective drag coefficient are derived from the balance of gravity and drag during steady descent. Landing detection combines low vertical speed with altitude stability.

Gases (3)

`air_quality_proxy, plume_detection, voc_profile.`

VOC/gas-resistance readings from the BME688 are turned into an air-quality proxy and a vertical VOC profile; plume detection flags localised concentration anomalies.

GNSS (4)

`displacement_from_start, ground_speed, ground_track_distance, vertical_speed.`

Horizontal metrics use great-circle (haversine) distance between fixes; vertical speed uses GNSS altitude deltas. All require valid GNSS fixes — with no fix, the analyses report unavailable rather than emitting zeros.

Magnetic (4)

`magnetic_anomaly, magnetic_field_magnitude, magnetic_heading, magnetic_inclination.`

Field magnitude is the Euclidean norm of the three magnetometer axes; heading and inclination are derived from the field vector; anomalies are deviations from the expected local field.

Power (3)

`battery_health, brownout_risk, voltage_trend.`

Voltage trend is a regression over the battery-voltage series; brownout risk compares projected voltage against a threshold; battery health summarises the discharge behaviour.

Radiation (4)

`dose_rate, mean_count_rate, radiation_altitude_correlation, radiation_burst.`

Count rates (CPM) are converted to an approximate dose rate; the altitude–radiation relationship is reported strictly as a correlation with its coefficient and sample count; bursts are transient rate excursions.

UV / Solar (2)

`uv_altitude_trend, uv_index.`

UV index from the UV sensor and its trend with altitude.

Composite (3)

`change_rate_monitor`, `correlation_matrix`, `environmental_hazard_index`.

Cross-sensor summaries: a rate-of-change monitor, a correlation matrix across key channels (correlation only — never causal), and a composite environmental hazard index.

Filtering and fusion

Before analyses run, the pipeline can apply configurable filters (stage 14) and sensor fusion (stage 15):

- **Kalman 1-D** (e.g. fused pressure), **moving average** (temperature), **EKF altitude** — all configured under `pipeline.filter_config`.
- Fused outputs are tagged `DataSource.FUSED`; filtered outputs are tagged appropriately so consumers always know the provenance of a number.

Narrative

`GET /api/v1/narrative` produces a natural-language description of the mission from the computed results. The narrative is generated from real analysis outputs and states uncertainty; it never asserts findings the analyses did not support.